TEMASEK JUNIOR COLLEGE

Preliminary Examinations
Higher 1

CANDIDATE NAME	
CIVICS GROUP	/
CENTER NUMBER	S
INDEX NUMBER	

CHEMISTRY 8872/02

Paper 2 **11th September 2012**

Candidates answer Section A on the Question Paper. **2 hours**

Additional Materials: Answer Paper
Data Booklet
Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your Civics Group number and candidate name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **two** questions on separate answer paper.

At the end of the examination, Section A and Section B will be handed in separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		
Paper 1		/30
Paper 2	A1	/
	A2	/
	A3	/
	A4	/
	B	/20
	B	/20
Total		/110

This document consists of **16** printed pages.

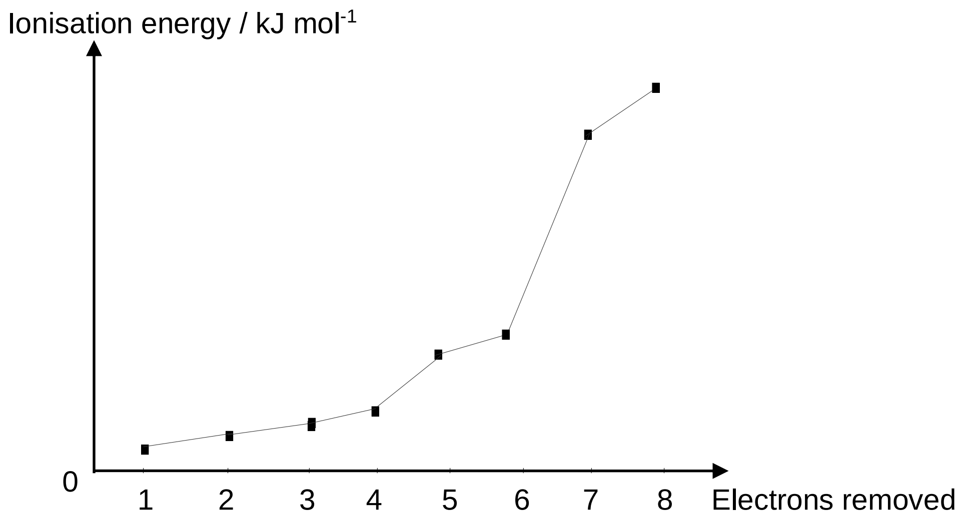
For Examiner's use

Section A

Answer **all** questions in this section in the spaces provided.

1 (a) **A**, **B**, **C**, **D** and **E** are a sequence of consecutive elements of increasing proton number in Periods 2 and 3.

The first eight ionisation energies of element **B** are shown in the sketch graph below.



(i) To which Group does element **B** belong to? Explain your answer.

....

....

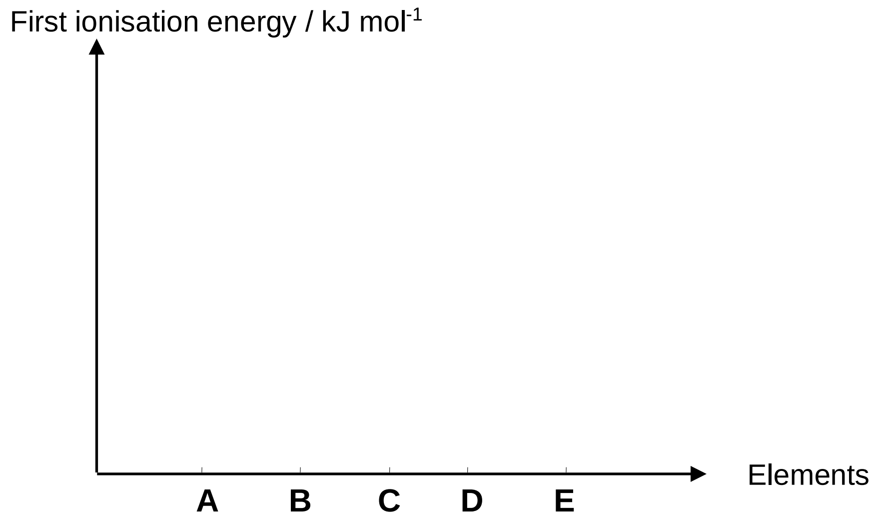
....

....

(ii) Draw and label the shape and symmetry of the atomic orbitals of the outer shell of element **B**.

For Examiner's use

(iii) Sketch the graph of first ionisation energies, in kJ mol^{-1}, against the elements **A** to **E** on the axes given.



(iv) Account for the sketch drawn.

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A, **B**, **C** and **E** form ions which are isoelectronic with atom **D**.

(v) Define the term 'isoelectronic'.

....

(vi) State the charges on the ions of **A**, **B**, **C** and **E**.

A: .... **B**: .... **C**: .... **E**: ....

[10]

(b) Halogens can form a variety of compounds with nitrogen and alkenes.

Nitrogen trifluoride, NF_3, is used to etch silicon in microelectronics. It can be prepared by reacting ammonia with fluorine. In this reaction, the fluorine oxidises the nitrogen in ammonia while the oxidation number of hydrogen is unchanged.

(i) Give a balanced equation for this reaction.

....

(ii) Predict the electrical conductivity of nitrogen trifluoride in the solid state.

....

....

(iii) Nitrogen trifluoride forms a compound with BCl_3. Describe the type of bond that is formed during this reaction. Draw a diagram to illustrate the shape of and bonding in the product.

For Examiner's use

(iv) Whereas nitrogen trifluoride is reasonably easy to handle, nitrogen trichloride is an extremely dangerous explosive. Suggest why nitrogen trifluoride is more stable than the other nitrogen trihalides.

[5]
[Total: 15]

For Examiner's use

2 **(a)** One of the main uses of ethane in the chemical industry is in the production of ethene which is used to make plastic. Ethane is produced during the decomposition of azomethane, CH_3NNCH_3, whereby nitrogen gas is formed as the other product.

$$CH_3N{=}NCH_3\ (g) \rightarrow CH_3CH_3\ (g) + N_2\ (g)$$

The table of bond energies is provided below.

bond	bond energy / kJ mol^{-1}
C – C	350
C – H	410
N – C	305
N – H	390
N = N	410
N ≡ N	994

(i) What do you understand by the term *bond energy*?

....

....

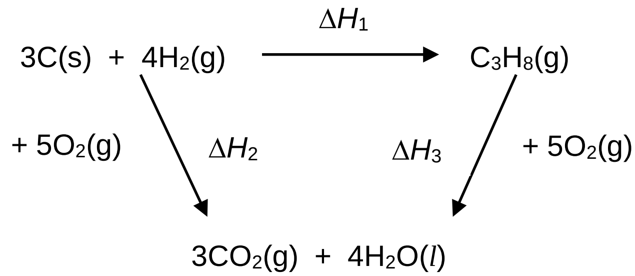
....

(ii) Use the table of bond energies above to calculate the enthalpy change for the reaction when azomethane decomposes to form ethane.

[4]

For Examiner's use

(b) The diagram below shows an energy cycle involving the fuel, propane.



(i) In the energy cycle above, what enthalpy change is represented by ΔH_1?

(ii) Write an equation that links ΔH_1, ΔH_2 and ΔH_3.

(iii) Use the following data to calculate the standard enthalpy change of combustion of propane.

$\Delta H^{\ominus}_{combustion}$ of carbon = -393 kJ mol^{-1}

$\Delta H^{\ominus}_{combustion}$ of hydrogen = -286 kJ mol^{-1}

$\Delta H^{\ominus}_{formation}$ of propane = -104 kJ mol^{-1}

[4]
[Total: 8]

For Examiner's use

3 Malachite is an ore of copper with the formula $Cu_x(CO_3)_y(OH)_z$. It has a vibrant colour and was used as a mineral pigment in green paint until the 18th century.

0.01 mol of a sample of malachite was dissolved in excess sulfuric acid and a gas was evolved.

Excess potassium iodide was then added to the resultant solution. The reaction of iodide and copper(II) ions produced a white precipitate, CuI and a brown iodine solution.

$$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$$

(a) The iodine produced required 0.02 mol of sodium thiosulfate for complete reaction.

(i) Write a balanced equation for the reaction between iodine and thiosulfate ions.

....

(ii) Hence determine the value of *x*.

[3]

(b) The gas evolved from the reaction of malachite with sulfuric acid was bubbled into excess calcium hydroxide. 0.01 mol of calcium carbonate was produced.

(i) Write a balanced equation for the reaction between the gas evolved and calcium hydroxide.

....

(ii) Hence determine the value of *y*.

[3]

For Examiner's use

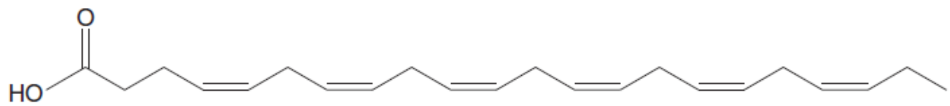
(c) Using your answers to **a(ii)** and **b(ii)**, deduce the value of z.

[1]
[Total: 7]

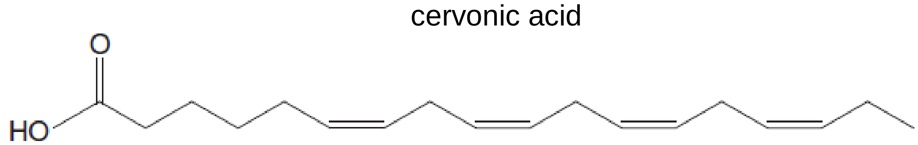
For Examiner's use

4 **(a)** Evidence is accumulating that omega-3 oils help to protect us from schizophrenia and depression, and even improve learning and memory. Omega-3 oils are glyceryl esters of omega-3 fatty acids.

Two omega-3 fatty acids are shown below.



cervonic acid



stearidonic acid

(i) Which configuration of the carbon-carbon double bonds is present in these molecules?

....

(ii) How many geometric isomers are there of cervonic acid, including the molecule shown?

....

(iii) What is the name of the intermolecular force that exists between the hydrocarbon chains?

....

[3]

(b) **(i)** It is possible to distinguish the different types of fatty acids and oils in the laboratory by measuring the degree of unsaturation in the hydrocarbon chains. Halogens react with alkenes in an addition reaction.

Name the product formed when propene reacts with liquid bromine.

....

For Examiner's use

(ii) The table below shows information about the two fatty acids.

Fatty acid	Molecular formula	Molar mass/g mol^{-1}
cervonic acid	$C_{22}H_{32}O_2$	328
stearidonic acid	$C_{18}H_{28}O_2$	276

The degree of unsaturation in a fatty acid is commonly expressed by the mass of iodine that reacts with 100 g of the acid. Use the information in the table above to calculate the mass of iodine that would react with 100 g of cervonic acid.

(iii) The interhalogen compound ICl also reacts with propene in an addition reaction. ICl reacts faster with propene than the halogens and so can be used to determine volumetrically the unsaturation in fatty acids and oils. Suggest why ICl reacts with propene faster than the pure halogens, Cl_2, Br_2 and I_2.

....

....

[4]

(c) The 'OXO' reaction (reaction **I** in the scheme below) is industrially important for making alcohols and aldehydes. For example, butan-1-ol and butanal can all be synthesised from propene, according to the following scheme.

$$\underset{\text{propene}}{CH_3CH{=}CH_2(g)} + CO(g) + H_2(g) \overset{\mathbf{I}}{\rightleftharpoons} \underset{\text{butanal}}{CH_3CH_2CH_2CHO} \xrightarrow{\mathbf{II}} \underset{\text{butan-1-ol}}{CH_3CH_2CH_2CH_2OH}$$

(i) State the reagents and conditions needed for reaction **II**.

....

For Examiner's use

(ii) State the reagents and conditions that will react with **both** butanal and butan-1-ol but **not** with propene and describe what would be observed.

Reagents and Conditions: ....

Observation: ....

....

[3]
[Total: 10]

Section B

Answer **two** of the three questions in this section on separate paper.

5 **(a)** Fuel engineers use the extent of the change of the reaction between carbon monoxide and steam to regulate the proportions of synthetic fuel mixtures.

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

(i) Write an expression for the K_c for this equilibrium.

(ii) In an experiment 0.250 mol of CO and 0.250 mol of steam were placed in a 125 cm^3 flask at 900 K and an equilibrium was allowed to be established. Given that 0.139 mol of CO_2 and 0.139 mol of H_2 were present in the equilibrium mixture, calculate K_c at this temperature.

(iii) When the temperature was lowered, more CO_2 and H_2 were formed. Explain, using Le Chatelier's principle, whether the forward reaction is exothermic or endothermic.

[5]

(b) State and explain how the acidities of water, ethanoic acid and ethanol compare with each other.

[4]

(c) RCO_2H is a weak monobasic acid which dissociates partially in solution.

$$RCO_2H + H_2O \rightleftharpoons RCO_2^- + H_3O^+$$

(i) What do you understand by the *Bronsted-Lowry* theory of acids and bases?

(ii) Identify the two acids and the two bases present in the equilibrium above.

[4]

(d) A solution of the weak acid, RCO_2H, was prepared by dissolving 2.41 g of RCO_2H in water and topping up to 250 cm^3 with deionized water in a volumetric flask. A 25.0 cm^3 sample of this solution was then pipetted into a conical flask and titrated with 0.150 mol dm^{-3} NaOH solution. 21.70 cm^3 of NaOH was required for a complete neutralisation.

(i) Suggest a suitable indicator for this titration.

(ii) Calculate the relative molecular mass of the acid and hence suggest its structural formula.

(iii) Before equivalence point was reached, an acidic buffer was formed. Explain, using appropriate equations, how it is able to resist changes in pH when a small amount of a base or an acid is added to it.

[7]
[Total: 20]

6 **(a)** The oxides of Na_2O, Al_2O_3 and SO_3 have the melting points of 1275 °C, 2072 °C and 17 °C respectively.

(i) Relate their melting points to their structures and bonding.

(ii) Describe their solubilities in, and reactions with, water. Give the approximate pH of any solution formed and write equations where appropriate.

(iii) Describe **two** reactions which demonstrate the amphoteric behaviour of Al_2O_3, writing equations where appropriate.

[9]

(b) Compound **S** has the molecular formula, $C_6H_{12}O_2$. When **S** is subjected to heating with aqueous HC*l*, compound **T**, $C_3H_6O_2$, and compound **U**, C_3H_8O, are formed. When **U** is added to hot acidified aqueous potassium manganate(VII), compound **W** is formed. Effervescence is observed when **T** reacts with aqueous sodium carbonate. **W** reacts with 2,4-dinitrophenylhydrazine but not Fehling's solution. Deduce the structures of **S**, **T**, **U** and **W**. Explain your answers clearly.

[8]

(c) Tartaric acid is a diprotic acid which occurs naturally in many plants, particularly grapes and bananas. The following shows the outline of a synthesis of tartaric acid:

$$CH_2(OH)CH{=}CHCH_2OH \xrightarrow{\textbf{I}} \textbf{J} \xrightarrow{\textbf{II}} \underset{\text{Tartaric acid}}{HO_2CCH(OH)CH(OH)CO_2H}$$

Suggest reagents and conditions for reactions **I** and **II**, and draw the structural formula of the intermediate **J**.

[3]

[Total: 20]

7 **(a)** A bromoalkane, RBr is hydrolysed by aqueous sodium hydroxide.

The following results were obtained from two experiments on such a hydrolysis. In each experiment, the overall [NaOH(aq)] remained virtually constant as indicated in the table.

Time/ min	[RBr]/ mol dm^{-3} when [OH$^-$] = 0.10 mol dm^{-3}	[RBr]/ mol dm^{-3} when [OH$^-$] = 0.15 mol dm^{-3}
0	0.0100	0.0100
40	0.0079	0.0070
80	0.0062	0.0049
120	0.0049	0.0034
160	0.0038	0.0024
200	0.0030	0.0017
240	0.0024	0.0012

Using the **same axes**, plot graphs for the **two** experiments. Use your graphs to answer the following questions. (*Show your working clearly.*)

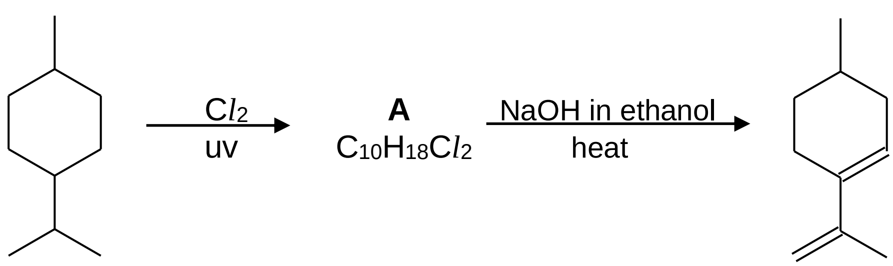
(i) Deduce the order with respect to the concentration of bromoalkane.

(ii) Deduce the order with respect to the concentration of sodium hydroxide.

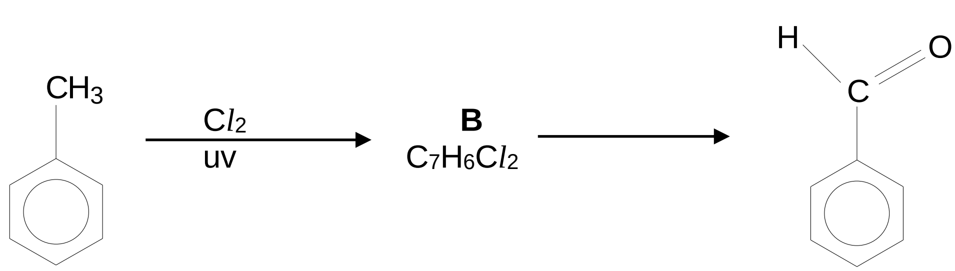
(iii) Construct a rate equation for the reaction and use it to calculate a value for the rate constant.

(iv) Use the concept of activation energy to explain how an increase in the concentration of sodium hydroxide increases the rate of reaction. [10]

(b) Consider the following schemes.



Only one product formed



benzaldehyde

Suggest a structural formula for each of the compounds **A** and **B** and state the reagents and conditions for the conversion of **B** to benzaldehyde.

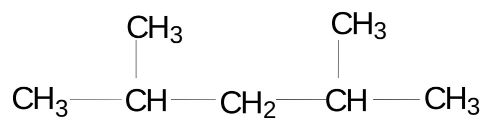
[3]

(c) An alkane **C**, C_7H_{16}, is a component of petrol. When **C** is treated with chlorine gas under free radical conditions, it produces a mixture of chlorinated compounds, including **D**, $C_7H_{14}Cl_2$. Upon elimination, **D** produces **E**, C_7H_{12}. When **E** is refluxed with hot acidic $KMnO_4$, three compounds, CO_2, $CH_3COCOCH_3$ and CH_3CO_2H, are formed in equimolar amounts.

(i) Suggest the structure for compound **E**, and hence deduce the structure of compound **C**.

(ii) Draw the four possible structures of **D**, $C_7H_{14}Cl_2$, that could give **E** on elimination.

(iii) Alkane **F**, another isomer of **C**, has the following structure.



F

Draw the structures of all the mono-chlorinated isomers that could be formed when **F** is treated with chlorine gas under free radical conditions. Suggest, with a reason, the approximate ratio in which they are formed.

[7]
[Total: 20]